

SOUTH HATTERAS BUOY 41002, USA

WMO: 992150

Lat: 31.887N

Lon: 74.921W

Elev: 0

StdP: 101.33

Time Zone: -5.00 (W05)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 9.3	(c) 10.8	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 17.9	(k) 15.9	(l) 16.8	(m) 15.1	(n) 9.7	(o) 350	(p) 0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 2.1	(c) 28.9	(d) N/A	(e) 28.6	(f) N/A	(g) 28.4	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 5.5	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 15.4	(b) 14.2	(c) 13.0	(e) 6.8	(f) 29.6	(g) 2.2	(h) 0.7	(i) 5.2	(j) 30.0	(k) 3.9	(l) 30.4	(m) 2.6	(n) 30.8	(o) 1.0	(p) 31.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.1	17.5	17.6	18.3	20.3	22.4	25.2	26.8	26.9	26.1	23.8	20.9	19.4		
	DBStd	4.02	3.18	2.93	2.82	1.78	1.45	1.40	0.94	0.91	1.12	1.83	2.39	2.83		
	HDD10.0	2	1	1	1	0	0	0	0	0	0	0	0	0		
	HDD18.3	165	53	42	34	5	0	0	0	0	0	1	7	23		
	CDD10.0	4427	234	213	258	309	386	455	521	522	483	429	327	291		
	CDD18.3	1548	28	21	33	64	127	205	262	264	233	171	84	55		
	CDH23.3	10418	3	0	2	18	195	1487	2772	2812	2125	852	131	23		
	CDH26.7	1318	0	0	0	0	0	87	472	509	225	26	0	0		
(14) Wind	WSAvg	6.8	8.1	8.0	8.0	7.0	6.3	5.7	6.0	5.3	5.9	6.5	7.2	7.4		
(15) Precipitation	PrecAvg	1447	140	131	121	85	85	106	123	131	155	127	122	121		
	PrecMax	2150	308	222	249	204	223	318	289	278	306	260	277	384		
	PrecMin	1094	50	37	33	29	29	13	21	48	34	36	44	36		
	PrecStd	203	61	47	55	42	46	68	77	53	77	60	51	67		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.9	23.0	23.6	24.7	26.1	28.6	29.4	29.3	29.0	28.7	26.1	24.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	23.0	22.6	23.0	23.9	25.5	28.3	28.9	28.9	28.5	27.4	25.2	24.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	22.3	22.1	22.5	23.1	24.9	27.7	28.5	28.6	28.2	26.8	24.7	23.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	DB	21.7	21.5	21.8	22.8	24.4	27.1	28.3	28.4	27.9	26.3	24.2	23.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.7	3.4	3.0	2.3	1.7	1.7	1.9	2.1	1.8	2.0	2.7	3.0		
		MCDBR	3.5	3.2	3.3	2.5	1.8	1.7	1.9	2.1	2.0	2.0	2.7	2.8		
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.361	0.370	0.382	0.407	0.439	0.451	0.459	0.448	0.420	0.394	0.368	0.364		
	taud		2.505	2.484	2.445	2.358	2.312	2.339	2.345	2.378	2.449	2.484	2.511	2.503		
	Ebn at Noon		859	887	899	885	855	840	832	838	851	849	844	834		
	Edh at Noon		88	98	109	124	131	127	126	120	108	97	87	85		
(44) All-Sky Solar Radiation	RadAvg		2.50	3.29	4.46	5.81	6.52	6.35	5.89	5.51	4.83	4.10	3.04	2.42		
	RadStd		0.27	0.40	0.42	0.39	0.39	0.47	0.62	0.46	0.36	0.28	0.23	0.21		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page